Experiment 009: Summary Report

Generated by run.py

1 Audit Data

This TeX file is intentionally a compact summary. The complete auditable record, including all bases, full differentials, boundary matrices, kernels, images, and representative coordinates, is stored in

`../data/chain_complex.json`.

2 Dimension Tables

Selected weight bound: $W = 8$. Resolution degree: 6.

Degree	Weight	Cell count
0	1	2
1	2	3
2	4	8
2	7	6
2	8	2
3	4	3
3	5	18
3	6	26
3	7	61
3	8	249
4	5	6
4	6	12
4	7	44
4	8	60
5	6	12
5	7	66
5	8	168
6	7	24
6	8	180

C_0	C_1	C_2	C_3	C_4	C_5	C_6
2	3	16	357	122	246	204

H_0	H_1	H_2	H_3	H_4	H_5
2	3	8	307	8	18

3 Key Matrices and Representatives

The full boundary matrices and representative coordinates are in the JSON audit file. Matrix rows are source basis elements and columns are target coordinates.

Boundary	Source dim	Target dim	Rank	Nonzero rows
∂_1	3	2	0	0
∂_2	16	3	0	0
∂_3	357	16	8	8
∂_4	122	357	42	42
∂_5	246	122	72	72
∂_6	204	246	156	156

3.1 Boundary Formula Summary

∂_1.

```
partial_1: all recorded formulas are zero.
```

∂_2.

```
partial_2: all recorded formulas are zero.
```

∂_3.

```
partial_3(s3_00103_w7) = s2_00009_w7
partial_3(s3_00104_w7) = s2_00010_w7
partial_3(s3_00105_w7) = s2_00011_w7
partial_3(s3_00106_w7) = s2_00012_w7
... 4 more nonzero formulas in chain_complex.json
```

∂_4.

```
partial_4(s4_00001_w5) = s3_00004_w5
partial_4(s4_00002_w5) = s3_00005_w5
partial_4(s4_00003_w5) = s3_00006_w5
partial_4(s4_00004_w5) = s3_00007_w5
... 38 more nonzero formulas in chain_complex.json
```

∂_5.

```
partial_5(s5_00001_w6) = s4_00007_w6
partial_5(s5_00002_w6) = s4_00008_w6
partial_5(s5_00003_w6) = s4_00009_w6
partial_5(s5_00004_w6) = s4_00010_w6
... 68 more nonzero formulas in chain_complex.json
```

∂_6.

```
partial_6(s6_00001_w7) = s5_00031_w7
partial_6(s6_00002_w7) = s5_00032_w7
partial_6(s6_00003_w7) = s5_00033_w7
partial_6(s6_00004_w7) = s5_00034_w7
... 152 more nonzero formulas in chain_complex.json
```

3.2 Representative Summary

H_0.

```
x
y
```

H_1.

```
r_xx
r_xy
r_yy
```

H_2.

```
s2_00001_w4
s2_00002_w4
s2_00003_w4
s2_00004_w4
s2_00005_w4
... 3 more representatives in chain_complex.json
```

H_3.

```
s3_00001_w4
s3_00002_w4
s3_00003_w4
s3_00010_w5
s3_00011_w5
... 302 more representatives in chain_complex.json
```

H_4.

```
s4_00043_w7
s4_00044_w7
s4_00075_w8
s4_00076_w8
s4_00077_w8
... 3 more representatives in chain_complex.json
```

H_5.

```
s5_00013_w7
s5_00014_w7
s5_00015_w7
s5_00016_w7
s5_00017_w7
... 13 more representatives in chain_complex.json
```

4 Weight-Bound Stability

The requested bounds $W = 6, 7, 8$ were compared. The selected output may be called stable only if the dimensions and representatives agree across these bounds.

Weight	H_0	H_1	H_2	H_3	H_4	H_5
6	2	3	8	41	0	0
7	2	3	8	78	2	18
8	2	3	8	307	8	18

Stability verdict: truncated evidence only; selected weight bounds disagree.